Fiscal Year:	FY 2025	**Task Last Updated:**	FY 04/11/2025
PI Name:	Contractor, Noshir Ph.D.		
Project Title:	Composing Teams with TEAMSTaR: Tool for Evaluating and Mitigating Space Team Risk		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	60208-0834	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2020 HERO 80JSC019N0001-TEAM: Team Composition-Appendix G
Start Date:	04/15/2021	**End Date:**	09/30/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	12	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/30/2025 per NSSC info and NASA-JSC (Ed., 9/10/24) NOTE: End date changed to 4/14/2025 per NSSC info via V. Lehman/JSC Grants Office (Ed., 4/17/24) NOTE: End date changed to 4/14/2024 per NSSC info via L. Barnes-Moten/JSC (Ed., 7/16/21)		
Key Personnel Changes/Previous PI:	In May 2024, Alina Lungeanu was moved from co-PI to Consultant status on TEAMSTAR.		
COI Name (Institution):	Bell, Suzanne Ph.D. (NASA Johnson Space Center) DeChurch, Leslie Ph.D. (Northwestern University, Evanston) Loerch, Linda M.S. (NASA Johnson Space Center)		
Grant/Contract No.:	80NSSC21K0925		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	As NASA sets its sight on more Earth-independent missions, such as Artemis missions to the Moon and on to Mars, team composition becomes a critical leverage point for mitigating risks. NASA has successfully designed crews with "the Right Stuff" for more than fifty years beginning with the Mercury and Apollo programs, then into the Shuttle period, and throughout the Skylab and International Space Station missions requiring highly specialized crews to live and work in space for extended periods of time. The key point of departure for deep space exploration is the complexity of missions and the autonomy with which the crew will work. Communication delays with support teams on Earth will necessitate that a relatively small crew take on greater responsibility for making critical decisions. This increased autonomy will occur despite the additional challenges posed by prolonged isolation and confinement and increased radiation exposure. Whereas the "Right Stuff" (Wolfe, 1979) emphasized the requisite individual characteristics, deep space missions also require the "Right Combination" of team members. With that as a backdrop, this project develops and validates TEAMSTaR (Tool for Evaluating And Mitigating Space Team Risks), a team composition decision support system, that can be used by stakeholders (e.g., schedule decision-makers) to predict how a hypothetical team's social relations are likely to evolve and influence crew performance over the course of a mission. The TEAMSTaR will enable decision-makers to evaluate composition scenarios for an entire set of teams, for single-member replacements, and/or for subsets of teams. To do this, we first leverage insights and data from recent NASA-funded team composition studies and thoughtfully refine and extend our agent-based models to include relevant input characteristics and their ability to predict team outcomes including team performance. We next conduct virtual experiments and gather stakeholder input to inform the development of TEAMSTaR, a team composition decision support system that utilizes insights from our updated agent-based models (ABMs) to enable real-time (or close to real-time) decision-making. Finally, we validate TEAMSTaR as a decision-making tool in short and long-term isolated, confined, and controlled environments. This project accomplishes five aims. Aim 1) Refine agent-based models looking at relevant input characteristics and their ability to predict team outcomes, including team performance. Aim 2) Identify and elaborate the scientific rationale for attributes used within the model, identifying factors known to affect crew functioning, crew member behavior, emergent characteristics that arise during team task completion. Aim 3) Develop and validate a Team Composition decision support system and user interface. Aim 4) Validate the refined model using a software prototype in at least one extended duration, isolated, and confined analog. Aim 5) Provide modeling and software prototypes that meet NASA Standard 7009a. With an eye toward the future of deep space exploration, this project leverages, and indeed advances, state of the art computational techniques to predict crew performance and to identify points of leverage in terms of team composition and task scheduling to optimize individual and team performance. Reference: Wolfe, T. (1979). The Right Stuff. New York: Farrar, Straus and Giroux.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	With an eye toward the future of deep space exploration, this project leverages and advances state of the art computational techniques to predict crew performance and to identify points of leverage in terms of team composition and task scheduling to optimize individual and team performance. Even though TEAMSTaR decision support system will be tested in space analog missions, it can be applied to teams operating on Earth in isolated and confined environments (ICE), such as expedition and science teams in the Arctic and Antarctic. The general framework of team composition and the analytic strategies developed in this project can be applied to Earth teams more generally.
Task Progress:	We have completed the fourth year of the project. Last year, we addressed Aims 1 and 2 by refining agent-based models and generating test case scenarios for crew and mission control. We also completed the design and development of the TEAMSTaR dashboard. During this past year, we completed data collection in the NASA Human Exploration Research Analog (HERA) Campaign 7 (C7) with the aim of validating the refined user interface and models (Aims 3 and 4). Aim 1 (refining the agent-based model looking at relevant input characteristics and their ability to predict team outcomes, including team performance) and Aim 2 (identifying and elaborating the scientific rationale for attributes used within the model). Last year, we finalized the agent-based model (ABM) and a list of attributes to predict team outcomes, as well as test case scenarios which were implemented in HERA C7. This year, the combined ABM and performance prediction model was implemented in data collection on a weekly basis, based on data collected pre-mission and in-mission from a collection of surveys. We used the ABMs to generate 22 sets of 4 crew networks (leadership, hindrance, positive affect, and shared mental models) from four unique crews. From each set of simulated networks, we generated two scenarios, containing multiple performance projections. Crews and ground support personnel (GSP) interacted with the performance projections on a weekly basis when performing the TEAMSTaR decision-making activity in-mission. Aim 3 (develop and validate TEAMSTaR, a Team Composition decision support system and user interface) and Aim 4 (validate the refined model using a software prototype in at least one extended-duration, isolated, and confined analog). We simulated crew relations and generated performance predictions on a weekly basis in-mission, and crews performed a series of three linked activities each week. First, the crew would perform a decision making activity using the TEAMSTaR dashboard, then the crew would perform a shared activity that was selected between two options, and then the crew would perform an operational task in pairs that were selected from the three possible arrangements. We collected data from four missions, with M3 being impacted such that data was not collected before MD13. We collected complete mission data from missions 1, 2, and 4. We collected user experience data, team outcomes, and performance data from the crews based on the activities they performed.
Bibliography Type:	Description: (Last Updated: 03/17/2026)

Abstracts for Journals and Proceedings	Chan M, DeChurch L, Contractor N. "PRISM: Performance Reimagined in Spaceflight Multiteam Systems." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Poster. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Abstracts for Journals and Proceedings	Chan M, Sudarsan S, DeChurch L, Lungeanu A, Bell S, Contractor N. "Paired for success: Operationally building crew communication." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Poster. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Abstracts for Journals and Proceedings	Contractor N, Chan M, Lungeanu A, DeChurch L, Bell S. "TEAMSTaR: Supporting resilient teams to go the distance." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Abstracts for Journals and Proceedings	Qian N, Chan M, Lungeanu A, DeChurch L, Bell S, Contractor N. "Improving space mission success through team state-building activities." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Poster. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Abstracts for Journals and Proceedings	Contractor N, Lungeanu A, DeChurch L, Chan M. "Teams that SIRIUSly go the distance: Effect of isolation and confinement on team performance." 75th International Astronautical Congress (IAC), Milan, Italy, Oct 14-18, 2024. IAC 2024 Congress Proceedings, 75th International Astronautical Congress (IAC), Milan, Italy, Oct 14-18, 2024. , Oct-2024
Abstracts for Journals and Proceedings	DeChurch L, Lungeanu A, Chan M, Contractor N. "Leading the crew to Mars: Evidence from NASA HERA analog crews." 75th International Astronautical Congress (IAC), Milan, Italy, Oct 14-18, 2024. IAC 2024 Congress Proceedings, 75th International Astronautical Congress (IAC), Milan, Italy, Oct 14-18, 2024. , Oct-2024
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Abstracts for Journals and Proceedings	Chan M, Contractor N, DeChurch L. "Leadership signatures in spaceflight multiteam systems." 2024 SIOP Annual Conference, Society for Industrial and Organizational Psychology, Chicago, IL, April 17-20, 2024. Abstracts. 2024 SIOP Annual Conference, Society for Industrial and Organizational Psychology, Chicago, IL, April 17-20, 2024. , Apr-2024
Articles in Peer-reviewed Journals	Contractor N, DeChurch L. "We are a go for launch! Working with NASA to forecast and improve team dynamics in space missions." J Appl Commun Res. 2025 Apr;53(1):62-5. https://doi.org/10.1080/00909882.2025.2472813 , Apr-2025